STAT 525 FA26 Project Guidelines

September 10, 2026

1 Overview

STAT 525 includes a group project involving analysis of real data. Students will work in groups of three or less.

Most important deliverables:

- **Group presentation:** Each group will present during the last week of classes, answer questions from the class, and participate in peer evaluation of other groups.
- **Individual final report:** Groups will collaborate on data analysis and create a shared pool of results. However, each student must write and submit **an independent final report**. The final report grade is individualized. Students may choose which results to include and how to interpret them; not all group results must be used.

It is strongly recommended that each group maintain a shared Google Doc (or similar platforms/software) to document progress and store results. If a student is reported to have contributed substantially less than expected, the instructor may request access to verify contributions using the revision history.

Each group is also recommended to meet with the TA or the instructor once before each key deadline (proposal, progress report, and presentation) to receive feedback and advices on the topic, data suitability, and analyses. This is not mandatory but early feedback could reduce the risk of an unexpected large workload at a very late time (for example, when no one realized that the data is actually poorly synthesized).

These guidelines may be updated. Any updates will be announced, and new changes will be marked in the updated file. Students are responsible for reviewing updates.

2 Timeline and required submissions

2.1 Step 1: Group formation and proposal

- **Group formation:** Students are expected to form groups of 3 classmates (or less) with shared interests. After forming a group, register it in Canvas (People → your session's Project Groups). Self sign-up is enabled.
- **Random assignment:** No deadline for grouping but students who have not joined a group by **September 24** may be randomly assigned to groups with empty slots by the instructor.
- **Proposal due: September 29**. The proposal is submitted once per group and must be at most **2 pages**. The proposal should include:
  - names of group members;
  - project goals and research question(s);
  - data source and a brief description of the dataset;
  - which variable(s) used as response, which as predictors, and their types (at least one continuous response variable, and some but not all predictors can be categorical);
  - why the question is interesting and worth studying.

The proposal may be reused in the final report with appropriate revisions. Groups are encouraged to make progress in parallel to the lecture/homework progress.

2.2 Step 2: Progress report

Due: November 12. The progress report is a draft (up to **8 pages**) and may be reused in the final report with revisions. It must contain two parts (up to **4 pages each**):

Part A: Data exploration and initial modeling The goal is to understand the data and prepare for multivariate linear regression. The draft should include:

- distributions of Y and each predictor (e.g., histograms/boxplots);
- scatterplots of Y versus quantitative predictors and appropriate visual summaries for categorical predictors;
- discussion of potential outliers or extreme values;
- assessment of whether transformations of Y or predictors may be needed (including nonlinear terms such as X^2 when appropriate);
- correlation analysis among quantitative predictors and discussion of potential multicollinearity;
- simple linear regressions of Y on meaningful predictors and a brief check of key assumptions; if issues are found, propose remedies (e.g., transformation, alternative specification, or careful treatment of unusual observations).

Part B: Deeper analysis and preliminary conclusions This section should summarize what has been done beyond exploratory summaries. It should include:

- analyses beyond simple linear regression (e.g., candidate multiple regression models);
- key statistical findings (tables/figures as needed) and subject-matter interpretations

2.3 Step 3: Group presentations

- All groups will present during the last week of classes.
- Presentation order will be determined randomly at the beginning of each session.
- The presentation is prepared jointly by the group.
- Time limit: **15 minutes** per group, including Q&A time.

2.4 Step 4: Individual final report

Due: December 23. Each student must submit an independent final report. More details below.

3 Expectations for group presentations

Groups should plan a clear, well-paced presentation. A suggested time allocation is:

- **3 minutes:** data context, data collection, and research question(s);
- **5 minutes:** exploratory analysis, model development, and main modeling choices;
- **5 minutes:** key results, interpretation, and conclusions.
- **2 minutes:** Q&A

Groups should address relevant items from the list below (not all items apply to every project):

- What question is the analysis intended to answer?
- How were the data collected, gathered, or sampled?
- Are confounders or strong associations among predictors present?
- Are interaction effects considered when appropriate?
- Is the final model reasonable and interpretable?
- What assumptions are being made, and how were they checked?
- What limitations arise from the data and modeling choices?

Presentation participation and peer evaluation Presentation participation and engagement (5% of the course grade) is earned by attentive participation, submission of peer evaluations, and constructive engagement during Q&A. Question quality is not graded. Peer evaluation will be based equally on:

- clarity of the project problem and goals;
- appropriateness of analysis choices;
- results and interpretation;
- overall presentation quality (organization, logical flow, readable slides, and clear tables/figures).

If the presentation reveals analytical gaps or major issues, the final report is expected to address them.

4 Expectations for the individual final report

4.1 Format and layout

- The first page must include the project title and the names of group members.
- Page limit: **12 pages** (excluding references and appendix materials). Use a readable font size (10–12pt recommended). Page numbers are required.
- Figures and tables must be numbered and have self-contained captions. Figures should be sized for readability and should not dominate the report. Only show key figures and tables and less important ones can be moved to appendix.
- Report numerical results in readable form (avoid scientific notation unless necessary). Key inference results should be stated in the text even if they also appear in a table (e.g., "$\hat{\beta} = 12.35$, $t = 2.78$, $p = 0.032$").
- The report should be polished and readable. Do not paste raw software output without formatting and explanation. The report file size has a 5MB limit.

The report must contain three main sections:

- **Introduction:** dataset background, data source, and project goals.
- **Methods and results:** exploratory summary, model building, diagnostics, and key findings (include only results used in the narrative).
- **Conclusion:** overall interpretation and take-home message. The conclusion should be understandable to a non-statistician.

4.2 Attachments: code and "AI" use

The use of large language models and generative models is encouraged but should be disclosed. Specific requirements related to such use are to be updated.

- R/python code must be submitted. Code should not appear in the main report body.
- Attachments may be included as an appendix (not counted toward the page limit), submitted as separate files.

4.3 Grading

Although the data analysis process and intermediate results may be shared within a group, the **final report is graded individually**. Each student is responsible for:

- selecting results that support the report narrative;
- providing correct statistical reasoning and interpretation;
- connecting conclusions to evidence from the analysis.

Students are strongly encouraged to keep brief notes on their individual contributions to the group work. Lack of contribution documentation may be considered in cases of reported contribution concerns.

Project grades are based primarily on clarity and correctness of statistical reasoning in the written report. Credit is given for appropriate use of methods covered in the course, justified treatment of influential observations, careful model building, validation of assumptions, correct interpretation, and clear communication. Deductions may be made for missing required components, unclear writing, interpretations not supported by the analysis, or modeling choices that are poorly justified.

Advanced methods beyond the course scope do not automatically receive extra credit; they are evaluated based on relevance and clarity. Any content that is identical to public sources and not disclosed will be investigated as a potential academic integrity violation.